Gram determinants on commuting schemes and the Chevalley restriction map

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Abstract

Let $\mathfrak{g}$ be a simple Lie algebra over an algebraically closed field of characteristic zero, $\mathfrak{t} \subset \mathfrak{g}$ a Cartan subalgebra with Weyl group W, and B an invariant form. Chen and Ngô conjectured that restriction to $\mathfrak{t}^d$ identifies the invariant functions on the scheme of commuting d-tuples in $\mathfrak{g}$ with W-invariant functions on $\mathfrak{t}^d$. We determine the SL_d-invariant part of degree $2d$ of this invariant ring: it is one-dimensional when d is smaller than the dual Coxeter number $h^\vee$, spanned by the Gram determinant $\det\big(B(x_i, x_j)\big)$, and zero for $d \geq h^\vee$. The proof transfers the Cachazo–Douglas–Seiberg–Witten theorem on the classical chiral ring of $\mathcal{N} = 1$ super-Yang–Mills theory, in which the gaugino bilinear S satisfies $S^{h^\vee - 1} \neq 0$ and $S^{h^\vee} = 0$, to commuting tuples. For $\operatorname{rk}\mathfrak{g} < d < h^\vee$ the Gram determinant is a nonzero nilpotent invariant with zero restriction to $\mathfrak{t}^d$, so the restriction map has a nonzero kernel. This range is nonempty for $\mathfrak{so}_n$ with $n \geq 7$ (smallest case $\mathfrak{so}_7$, $d = 4$) and for every exceptional type (smallest case $\mathfrak{g}_2$, $d = 3$), and empty in types A and C, where the conjecture is a theorem. The case $\mathfrak{so}_7$, $d = 4$ contradicts a published theorem of Song, Xia and Xu; we show that their Lemma 5.2 fails for every $d \geq 2$ and locate the step of its proof that breaks. In the super-commuting setting of Chang, the same component is an irreducible $\mathfrak{gl}(3|2)$-module of the classical Q-cohomology of $\mathcal{N} = 4$ super-Yang–Mills theory whose purely odd vector is S^d; for $\operatorname{rk}\mathfrak{g} < d < h^\vee$ these gaugino powers are the non-Coulomb monotone classes of Kibe, van Leuven and Mathieson, and the Gram determinant is their counterpart on the purely even commuting scheme. With the count of Chang and Lin, the $\mathfrak{so}_7$ component at $d = 4$ contains the non-Cartan class in the first classical mismatch between $\mathfrak{so}_7$ and $\mathfrak{sp}_6$.

1 Introduction

Let k be an algebraically closed field of characteristic zero, G a reductive group over k with Lie algebra $\mathfrak{g}$, $\mathfrak{t} \subset \mathfrak{g}$ a Cartan subalgebra and $W = N_G(\mathfrak{t})/Z_G(\mathfrak{t})$. For $d \geq 1$ the commuting scheme $\mathcal{C}_{\mathfrak{g}}^d \subset \mathfrak{g}^d$ is the scheme-theoretic zero fibre of the commutator map

$$\mathfrak{g}^d \longrightarrow \prod_{i<j} \mathfrak{g}, \qquad (x_1, \ldots, x_d) \longmapsto \big([x_i, x_j]\big)_{i<j},$$

so $k[\mathcal{C}_{\mathfrak{g}}^d] = k[\mathfrak{g}^d]/I_d$, where I_d is generated by the coordinate functions of all $[x_i, x_j]$. Restriction along $\mathfrak{t}^d \subset \mathcal{C}_{\mathfrak{g}}^d$ defines

$$\Phi : \ k[\mathcal{C}_{\mathfrak{g}}^d]^G \longrightarrow k[\mathfrak{t}^d]^W.$$

For $d = 1$ this is the Chevalley restriction isomorphism. Chen and Ngô [8, Conj. 3.1] conjectured that Φ is an isomorphism for every reductive G and every d, equivalently that $\mathcal{C}_{\mathfrak{g}}^d /\!\!/ G$ is reduced and normal; their weaker Conjecture 3.2 (existence of a spectral data morphism) already forces Φ to be injective [8, §3]. The conjecture is a theorem for GL_n (Vaccarino [24], Domokos [11];

Chen and Ngô prove Conjecture 3.2 for GL_n in [8, Thm. 3.3]) and for Sp_{2n} (Chen–Ngô [9, Thm. 4.1]); Song, Xia and Xu [23, Thm. 1.2] state it for O_n and SO_n. For $d = 2$ it follows for every connected reductive group from Li–Nadler–Yun [19, Thm. 1.4.11], stated there under their Ansatz 1.4.2, and from the normality of $\mathcal{C}^2_{\mathfrak{g}}$ proved by Sempliner and Yang [22] together with [8, §3]; for $\mathfrak{g}_2$ see also Goyal [15]. For reduced schemes, $\mathfrak{t}^d/\!/W \to (\mathcal{C}^d_{\mathfrak{g}})_{\mathrm{red}}/\!/G$ is an isomorphism for types A, B, C, D and G_2 by Hunziker [16], and for every reductive G it is finite and bijective on k-points [16, Thms. 6.2, 6.3], building on Richardson [20].

We study the Gram determinant. Fix a nondegenerate invariant symmetric form B on $\mathfrak{g}$ and set

$$T(x) = \big(B(x_i, x_j)\big)_{1\le i,j\le d}, \qquad \det T \in k[\mathfrak{g}^d]^G_{2d}.$$

Under GL_d acting on $(x_1, \dots, x_d)$, $\det T$ transforms by $\det^2$, so it lies in the SL_d-invariant part of degree $2d$. When $d > \operatorname{rk}\mathfrak{g}$, any d vectors of $\mathfrak{t}$ are linearly dependent and $\det T|_{\mathfrak{t}^d} = 0$. The question is whether $\det T$ lies in I_d.

The answer comes from the classical chiral ring of four-dimensional $\mathcal{N} = 1$ super-Yang–Mills theory. Cachazo, Douglas, Seiberg and Witten [3] considered the supercommutative algebra A generated by two odd copies X, Y of $\mathfrak{g}$ subject to $\{X, X\} = \{X, Y\} = \{Y, Y\} = 0$, and the invariant $S = B(X, Y)$. Their conjecture, proved for every simple $\mathfrak{g}$ [3, 25, 12, 13, 18, 4], states that $A^{\mathfrak{g}}$ is generated by S, that $S^{h^\vee} = 0$ and that $S^{h^\vee - 1} \neq 0$, where $h^\vee$ is the dual Coxeter number. The substitution $x_i = Xu_i + Yv_i$, with odd auxiliary variables u_i, v_i, produces commuting d-tuples on which $\det T$ takes the value $(-1)^d(d+1)!\, S^d \prod_i u_i v_i$. Super Schur–Weyl duality identifies the SL_d-invariant part of $k[\mathcal{C}^d_{\mathfrak{g}}]^G$ of degree $2d$ with the SL_2-invariant part of $A^{\mathfrak{g}}$ of degree $2d$, which is spanned by S^d.

Theorem A (Theorem 2.3)**.** Let $\mathfrak{g}$ be simple and $d \ge 1$. The SL_d-invariant subspace of $k[\mathcal{C}^d_{\mathfrak{g}}]^G$ of degree $2d$ has dimension 1 if $d < h^\vee$ and 0 if $d \ge h^\vee$. For $d < h^\vee$ it is spanned by the class of $\det T$.

Corollary B (Corollary 2.4)**.** If $\operatorname{rk}\mathfrak{g} < d < h^\vee$, then the class of $\det T$ is a nonzero nilpotent element of $k[\mathcal{C}^d_{\mathfrak{g}}]^G$ with $\Phi(\det T) = 0$. Hence $\ker\Phi \neq 0$, the nilradical of $k[\mathcal{C}^d_{\mathfrak{g}}]^G$ is nonzero, and Conjectures 3.1 and 3.2 of [8] fail for $(\mathfrak{g}, d)$.

The range $\operatorname{rk}\mathfrak{g} < d < h^\vee$ is empty for types A and C, in agreement with the theorems of Vaccarino, Domokos and Chen–Ngô, and nonempty for B_n $(n \ge 3)$, D_n $(n \ge 4)$ and every exceptional type (Table 1). The smallest case is $\mathfrak{g}_2$ with $d = 3$; there $\det T$ spans $\ker\Phi$ in all degrees at most 10, and $(\det T)^2 \in I_3$ (Example 6.1).

Corollary C (Corollary 7.4)**.** For $n \ge 7$, [23, Thm. 1.2(1), Cor. 1.3] fail for $G = O_n$ and $G = SO_n$; the smallest case is $n = 7$, $d = 4$. [23, Lemma 5.2] fails for every $d \ge 2$.

Section 7 proves the statement about [23, Lemma 5.2] directly: on the hypersurface where that lemma looks for solutions, the $2d \times 2d$ matrix it uses has rank at most 3. Section 8 treats the super version of Theorem A, which concerns the classical Q-cohomology of $\mathcal{N} = 4$ super-Yang–Mills theory.[1]

Kibe, van Leuven and Mathieson [17, eq. (1)] identified the powers S^m, $\operatorname{rk}\mathfrak{g} < m < h^\vee$, of the gaugino bilinear as non-Coulomb monotone classes of this Q-cohomology, nonzero by External input 2.2; they check the nonvanishing directly for $\mathfrak{so}_7, \dots, \mathfrak{so}_{12}$ [17, §4.5], relate S^4 for $\mathfrak{so}_7$ to the class of [6, 10], and construct a fortuitous partner of S^4 under the one-loop supercharge [17, §5]. Their classes are the purely odd vectors of the modules of Corollary 8.1

[1] By [7] and [5, Prop. 4], the classical Q-cohomology of top degree is the invariant ring of a super-commuting scheme with three even and two odd directions, and its purely even part is $k[\mathcal{C}^3_{\mathfrak{g}}]^G$. For $\mathfrak{g}_2$ the class of $\det T$ lies in Chang's multidegree $n = (0, 0, 2, 2, 2)$ (charges $(J_1, J_2, q_1, q_2, q_3) = (0, 0, 2, 2, 2)$ in the conventions of [5, §5.2]) and has zero Cartan restriction.

(Remark 8.3(d)). The present note adds the passage to the purely even commuting scheme: by Theorem A the gaugino power S^d on $\mathcal{C}_{\Pi k^2}$ and the Gram determinant on $\mathcal{C}_{\mathfrak{g}}^d$ have the same multiplicity space L_d. This gives Corollaries B and C, the full $\mathbb{S}_{(2^d)}$-component for every E (Corollary 8.1), and injectivity of the super Cartan restriction for $\mathfrak{sp}_{2N}$ (Corollary 8.2).

2 Setting and statements

Definition 2.1 (Commuting superschemes)**.** Let E be a finite-dimensional super vector space over k. The space $\mathrm{Hom}(E,\mathfrak{g})$ has coordinate superalgebra $\mathrm{Sym}(E\otimes\mathfrak{g}^*)$, the supersymmetric algebra. For $e,e'\in E$ and $\xi\in\mathfrak{g}^*$ let $\rho_E(e\wedge e'\otimes\xi)\in\mathrm{Sym}^2(E\otimes\mathfrak{g}^*)$ be the function $\theta\mapsto\xi([\theta(e),\theta(e')])$; it depends on $e\wedge e'$ in the super exterior square Λ^2E. Set

$$\mathcal{O}(\mathcal{C}_E)=\mathrm{Sym}(E\otimes\mathfrak{g}^*)/J_E,\qquad J_E=\big(\rho_E(\Lambda^2E\otimes\mathfrak{g}^*)\big).$$

Its points over a supercommutative k-algebra R are the even elements θ of $\mathrm{Hom}(E,\mathfrak{g})\otimes R$ with $[\theta(e),\theta(e')]=0$ for all $e,e'\in E$. The group $\mathrm{GL}(E)\times G$ acts. For $E=k^d$ this is the commuting scheme, $\mathcal{O}(\mathcal{C}_{k^d})=k[\mathcal{C}_{\mathfrak{g}}^d]$ [8, §3]; for $E=k^{d_B|d_F}$ it is the super-commuting scheme $\mathcal{C}_{\mathfrak{g}}^{d_B|d_F}$ of Chang [5, §2]. The Cartan restriction is

$$\Phi_E:\mathcal{O}(\mathcal{C}_E)^G\longrightarrow\mathcal{O}(\mathrm{Hom}(E,\mathfrak{t}))^W.$$

External input 2.2 (CDSW conjecture. Part (i) for every simple $\mathfrak{g}$: Kumar [18]. Parts (i)–(iii) for classical $\mathfrak{g}$: [3, 25] (for (iii) in types B and D, [25] uses instantons), algebraically [13]; for $\mathfrak{g}_2$: [12]; for F_4,E_6,E_7,E_8: [4, Prop. 2, Thm. 1, Cor. 2])**.** Let $\mathfrak{g}$ be simple with dual Coxeter number $h^\vee$, (e_a) a basis of $\mathfrak{g}$ and X^a,Y^a odd generators. Put $X=\sum_a e_a\otimes X^a$, $Y=\sum_a e_a\otimes Y^a$, and let A be the quotient of $\Lambda(X^a,Y^a)$ by the coordinates of

$$\{X,X\}=\{X,Y\}=\{Y,Y\}=0,\qquad\text{where}\quad\{X,Y\}:=\textstyle\sum_{a,b}[e_a,e_b]\otimes X^aY^b$$

and similarly for $\{X,X\}$, $\{Y,Y\}$; equivalently, $A=\Lambda(\mathfrak{g}\oplus\mathfrak{g})/I$ with I generated by the copies of $\mathfrak{g}$ in bidegrees $(2,0)$, $(1,1)$, $(0,2)$ [12, §1]. Let $S=\sum_{a,b}B(e_a,e_b)X^aY^b$. Then [12, Conj. 1.1]

$$\text{(i)}\ \ A^{\mathfrak{g}}\text{ is generated by }S,\qquad\text{(ii)}\ \ S^{h^\vee}=0,\qquad\text{(iii)}\ \ S^{h^\vee-1}\neq0.$$

The algebra A, the element S and the action of G are defined over $\mathbb{Q}$, so (i)–(iii) hold over every field of characteristic zero. In the notation of Definition 2.1, $A=\mathcal{O}(\mathcal{C}_{\Pi k^2})$, where Πk^2 is the purely odd space of dimension $0|2$ and $\mathfrak{g}^*$ is identified with $\mathfrak{g}$ through B.

Theorem 2.3 (Gram component)**.** *Let $\mathfrak{g}$ be simple, G connected and $d\geq1$, and let $\mathbb{S}_\lambda$ be the super Schur functor (External input 4.2). There is a vector space L_d, independent of E, such that for every finite-dimensional super vector space E the $\mathbb{S}_{(2^d)}(E)$-isotypic component of the $\mathrm{GL}(E)$-module $\mathcal{O}(\mathcal{C}_E)^G_{2d}$ is*

$$\mathcal{O}(\mathcal{C}_E)^G_{(2^d)}\ \cong\ \mathbb{S}_{(2^d)}(E)\otimes L_d,\qquad\dim L_d=\begin{cases}1, & d<h^\vee,\\ 0, & d\geq h^\vee,\end{cases}$$

where $(2^d)=(2,\dots,2)$ has d parts. For $E=k^d$ this component is the SL_d-invariant subspace of degree $2d$ of $k[\mathcal{C}_{\mathfrak{g}}^d]^G$, and for $d<h^\vee$ it is spanned by the class of $\det T$.

Corollary 2.4. *Let $\mathfrak{g}$ be simple with $\mathrm{rk}\,\mathfrak{g}<d<h^\vee$. Then the class of $\det T$ in $k[\mathcal{C}_{\mathfrak{g}}^d]^G$ is nonzero and nilpotent, and $\Phi(\det T)=0$. Consequently $\ker\Phi\neq0$, the nilradical of $k[\mathcal{C}_{\mathfrak{g}}^d]^G$ is nonzero, and Conjectures 3.1 and 3.2 of [8] fail for $(\mathfrak{g},d)$.*

type	rk	$h^\vee$	d with $\mathrm{rk} < d < h^\vee$
A_n	n	$n+1$	–
B_n, $n \geq 2$	n	$2n-1$	$n+1 \leq d \leq 2n-2 \quad (n \geq 3)$
C_n, $n \geq 3$	n	$n+1$	–
D_n, $n \geq 4$	n	$2n-2$	$n+1 \leq d \leq 2n-3$
G_2	2	4	$d = 3$
F_4	4	9	$5 \leq d \leq 8$
E_6	6	12	$7 \leq d \leq 11$
E_7	7	18	$8 \leq d \leq 17$
E_8	8	30	$9 \leq d \leq 29$

Table 1: Pairs $(\mathfrak{g}, d)$ covered by Corollary 2.4. For $\mathfrak{so}_n$: $n = 2m+1 \geq 7$ with $m+1 \leq d \leq 2m-2$, and $n = 2m \geq 8$ with $m+1 \leq d \leq 2m-3$.

3 The transfer

Lemma 3.1 (Grassmann Gram determinant)**.** *Let $\Lambda = \Lambda[u_1, \dots, u_d, v_1, \dots, v_d]$ be the exterior algebra on odd generators and $N_{ij} = u_i v_j + u_j v_i \in \Lambda_{\bar{0}}$. Then*

$$\det(N_{ij})_{1 \leq i,j \leq d} = (d+1)!\; u_1 v_1 u_2 v_2 \cdots u_d v_d.$$

Proof. Adjoin odd variables $\theta_i, \bar{\theta}_i$ and let $\int$ extract the coefficient of $\Theta = \bar{\theta}_1 \theta_1 \cdots \bar{\theta}_d \theta_d$. For a matrix M with entries in the commutative ring $\Lambda_{\bar{0}}$, the top component of $\exp(\sum_{i,j} \bar{\theta}_i M_{ij} \theta_j)$ is

$$\tfrac{1}{d!} \Big(\sum_{i,j} \bar{\theta}_i M_{ij} \theta_j \Big)^d = \sum_{\sigma \in \mathfrak{S}_d} \prod_r M_{r\sigma(r)}\, \bar{\theta}_r \theta_{\sigma(r)} = \det M \; \Theta,$$

since reordering the odd θ's among the even pairs $\bar{\theta}_r \theta_{\sigma(r)}$ produces $\operatorname{sgn} \sigma$. Hence $\det M = \int \exp(\bar{\theta} M \theta)$. For $M = N$, with the even elements

$$\alpha = \sum_i \bar{\theta}_i u_i, \quad \beta = \sum_j v_j \theta_j, \quad \gamma = \sum_i \bar{\theta}_i v_i, \quad \varepsilon = \sum_j u_j \theta_j,$$

one has $\bar{\theta}_i u_j v_i \theta_j = -\bar{\theta}_i v_i\, u_j \theta_j$ and therefore $\sum_{i,j} \bar{\theta}_i N_{ij} \theta_j = \alpha\beta - \gamma\varepsilon$. As $\alpha, \beta, \gamma, \varepsilon$ commute,

$$\det N = \int \sum_{r+s=d} \frac{(-1)^s}{r!\, s!}\, \alpha^r \gamma^s\, \beta^r \varepsilon^s.$$

Expanding the powers,

$$\alpha^r \gamma^s = r!\, s! \sum_{I \sqcup J = [d],\, |I| = r} \prod_{i \in I} \bar{\theta}_i u_i \prod_{j \in J} \bar{\theta}_j v_j, \qquad \beta^r \varepsilon^s = r!\, s! \sum_{K \sqcup L = [d],\, |K| = r} \prod_{i \in K} v_i \theta_i \prod_{j \in L} u_j \theta_j.$$

A product is nonzero exactly when every u_m and every v_m occurs once, i.e. when $L = J$ and $K = I$. Then

$$\bar{\theta}_i u_i\, v_i \theta_i = \bar{\theta}_i \theta_i\, u_i v_i, \qquad \bar{\theta}_j v_j\, u_j \theta_j = -\bar{\theta}_j \theta_j\, u_j v_j,$$

so each surviving term equals $(-1)^s\, \Theta \prod_m u_m v_m$. Collecting the coefficients,

$$\det N = \sum_{r=0}^{d} \frac{(-1)^s}{r!\, s!}\, (r!\, s!)^2 \binom{d}{r} (-1)^s \prod_m u_m v_m = \sum_{r=0}^{d} d! \prod_m u_m v_m = (d+1)! \prod_m u_m v_m.$$

□

Proposition 3.2 (Transfer)**.** *Let $\mathfrak{g}$ be simple, A as in External input 2.2, Λ as in Lemma 3.1, and $R_d = (A \otimes \Lambda)_{\bar{0}}$, a commutative k-algebra. Put*

$$x_i = X u_i + Y v_i = \sum_a e_a \otimes \big(X^a u_i + Y^a v_i\big) \in \mathfrak{g} \otimes R_d, \qquad 1 \le i \le d.$$

Then:

(i) $[x_i, x_j] = 0$ *for all* i, j*, so* $(x_1, \dots, x_d) \in \mathcal{C}_{\mathfrak{g}}^d(R_d)$*;*

(ii) $B(x_i, x_j) = -S\,(u_i v_j + u_j v_i)$*;*

(iii) the homomorphism $\varphi : k[\mathcal{C}_{\mathfrak{g}}^d] \to R_d$ *defined by* $(x_1, \dots, x_d)$ *satisfies* $\varphi(\det T) = (-1)^d (d+1)!\, S^d \otimes u_1 v_1 \cdots u_d v_d$*.*

Proof. (i) Moving the odd u_i or v_i past one odd generator,

$$[x_i, x_j] = -\sum_{a,b} [e_a, e_b] \otimes \big(X^a X^b\, u_i u_j + X^a Y^b\, u_i v_j + Y^a X^b\, v_i u_j + Y^a Y^b\, v_i v_j\big).$$

The four coefficients are $\{X,X\}$, $\{X,Y\}$, $\{Y,X\} = \{X,Y\}$ and $\{Y,Y\}$. For instance, in $U(\mathfrak{g}) \otimes A$ one has $XX = \sum_{a,b} e_a e_b X^a X^b = \frac{1}{2} \sum_{a,b} [e_a, e_b] X^a X^b$ (antisymmetrize in a, b using $X^b X^a = -X^a X^b$), so $XX + XX = \sum_{a,b} [e_a, e_b] X^a X^b$; similarly $\sum_{a,b} [e_a, e_b] Y^a X^b = \sum_{a,b} [e_a, e_b] X^a Y^b$. All four vanish in A.
(ii) With $B_{ab} = B(e_a, e_b)$,

$$B(x_i, x_j) = -\sum_{a,b} B_{ab} \big(X^a X^b\, u_i u_j + X^a Y^b\, u_i v_j + Y^a X^b\, v_i u_j + Y^a Y^b\, v_i v_j\big).$$

Symmetry of B_{ab} and antisymmetry of $X^a X^b$, $Y^a Y^b$ remove the first and last terms; $\sum B_{ab} Y^a X^b = -\sum B_{ab} X^b Y^a = -S$ and $v_i u_j = -u_j v_i$ give $B(x_i, x_j) = -S\, u_i v_j - S\, u_j v_i$.
(iii) By (i) the evaluation $k[\mathfrak{g}^d] \to R_d$ kills I_d. By (ii) and Lemma 3.1, with S even,

$$\varphi(\det T) = \det\big(-S\, N_{ij}\big) = (-S)^d \det N = (-1)^d (d+1)!\, S^d \otimes u_1 v_1 \cdots u_d v_d.$$

□

Corollary 3.3. *If $d < h^\vee$, then $\det T \notin I_d$.*

Proof. By External input 2.2(iii), $S^d \neq 0$ in A for $d \le h^\vee - 1$. The algebra $A \otimes \Lambda$ is free over A on the monomials in u, v, so $S^d \otimes u_1 v_1 \cdots u_d v_d \neq 0$ and Proposition 3.2(iii) gives $\varphi(\det T) \neq 0$. Since φ factors through $k[\mathcal{C}_{\mathfrak{g}}^d] = k[\mathfrak{g}^d]/I_d$, the class of $\det T$ is nonzero. □

4 The Gram component

Lemma 4.1 (One $\mathfrak{S}_n$-module for all E)**.** *For each $n \ge 0$ there is a $G \times \mathfrak{S}_n$-module V_n, independent of E, with natural isomorphisms*

$$\mathcal{O}(\mathcal{C}_E)_n \cong \big(E^{\otimes n} \otimes V_n\big)_{\mathfrak{S}_n},$$

where $\mathfrak{S}_n$ permutes the tensor factors of $E^{\otimes n}$ with Koszul signs and those of V_n.

Proof. The supersymmetric power is $\mathrm{Sym}^n(E\otimes\mathfrak{g}^*)=\big(E^{\otimes n}\otimes(\mathfrak{g}^*)^{\otimes n}\big)_{\mathfrak{S}_n}$. Let $c:\mathfrak{g}^*\to(\mathfrak{g}^*)^{\otimes 2}$ be the transpose of the bracket and $K_2=c(\mathfrak{g}^*)$, a space of antisymmetric tensors. The function $\rho_E(e\wedge e'\otimes\xi)$ is the class of $e\otimes e'\otimes c(\xi)$, so the degree-$n$ part of J_E is the image of $E^{\otimes n}\otimes\big(K_2\otimes(\mathfrak{g}^*)^{\otimes(n-2)}\big)$, which equals $\big(E^{\otimes n}\otimes K_n\big)_{\mathfrak{S}_n}$ for the $\mathfrak{S}_n$-submodule $K_n\subset(\mathfrak{g}^*)^{\otimes n}$ generated by $K_2\otimes(\mathfrak{g}^*)^{\otimes(n-2)}$. The functor of $\mathfrak{S}_n$-coinvariants is exact in characteristic zero, hence $\mathcal{O}(\mathcal{C}_E)_n=(E^{\otimes n}\otimes V_n)_{\mathfrak{S}_n}$ with $V_n=(\mathfrak{g}^*)^{\otimes n}/K_n$. □

External input 4.2 (Super Schur–Weyl duality, Sergeev [21], Berele–Regev [1])**.** For a partition $\lambda\vdash n$ with Specht module Sp_λ, the super Schur functor is $\mathbb{S}_\lambda(E)=(E^{\otimes n}\otimes\mathrm{Sp}_\lambda)_{\mathfrak{S}_n}$. For $E=k^{m|p}$, $\mathbb{S}_\lambda(E)\neq 0$ if and only if $\lambda_{m+1}\le p$, and the nonzero $\mathbb{S}_\lambda(E)$ are irreducible and pairwise non-isomorphic $\mathrm{GL}(E)$-modules. In particular $\mathbb{S}_\lambda(k^d)$ is the irreducible GL_d-module of highest weight λ (zero if λ has more than d parts), $\mathbb{S}_\lambda(\Pi k^e)\cong\mathbb{S}_{\lambda'}(k^e)$ as GL_e-modules with λ' the transpose partition, and $\mathrm{Sym}(E\otimes F)=\bigoplus_\lambda\mathbb{S}_\lambda(E)\otimes\mathbb{S}_\lambda(F)$ for F purely even.

Decompose $V_n=\bigoplus_{\lambda\vdash n}\mathrm{Sp}_\lambda\otimes M_\lambda$ with $M_\lambda=\mathrm{Hom}_{\mathfrak{S}_n}(\mathrm{Sp}_\lambda,V_n)$, a G-module. Lemma 4.1 and External input 4.2 give the $\mathrm{GL}(E)\times G$-decomposition

$$\mathcal{O}(\mathcal{C}_E)_n=\bigoplus_{\lambda\vdash n}\mathbb{S}_\lambda(E)\otimes M_\lambda,\qquad \mathcal{O}(\mathcal{C}_E)_n^G=\bigoplus_{\lambda\vdash n}\mathbb{S}_\lambda(E)\otimes M_\lambda^G. \tag{1}$$

Lemma 4.3 (Naturality of the Cartan restriction)**.** *For each $\lambda\vdash n$ there is a linear map $\phi_\lambda:M_\lambda^G\to\mathbb{S}_\lambda(\mathfrak{t}^*)^W$, independent of E, with*

$$\Phi_E=\bigoplus_\lambda \mathrm{id}_{\mathbb{S}_\lambda(E)}\otimes\phi_\lambda.$$

In particular Φ_E vanishes on $\mathbb{S}_\lambda(E)\otimes M_\lambda^G$ when λ has more than $\operatorname{rk}\mathfrak{g}$ parts.

Proof. Restriction $\mathfrak{g}^*\to\mathfrak{t}^*$ induces a $\mathfrak{S}_n$-equivariant map $(\mathfrak{g}^*)^{\otimes n}\to(\mathfrak{t}^*)^{\otimes n}$, which kills K_n because $\mathfrak{t}$ is abelian; let $V_n\to(\mathfrak{t}^*)^{\otimes n}$ be the induced map. Applying $(E^{\otimes n}\otimes-)_{\mathfrak{S}_n}$ gives the restriction $\mathcal{O}(\mathcal{C}_E)_n\to\mathrm{Sym}^n(E\otimes\mathfrak{t}^*)$, and $\mathrm{Hom}_{\mathfrak{S}_n}(\mathrm{Sp}_\lambda,(\mathfrak{t}^*)^{\otimes n})=\mathbb{S}_\lambda(\mathfrak{t}^*)$ by External input 4.2. Taking G-invariants on the source, whose image is W-invariant, gives ϕ_λ. Finally $\mathbb{S}_\lambda(\mathfrak{t}^*)=0$ when λ has more than $\dim\mathfrak{t}$ parts. □

Proof of Theorem 2.3. By (1) and External input 4.2, the $\mathbb{S}_{(2^d)}(E)$-isotypic component of $\mathcal{O}(\mathcal{C}_E)_{2d}^G$ is $\mathbb{S}_{(2^d)}(E)\otimes L_d$ with $L_d=M_{(2^d)}^G$, which is independent of E.

Odd side. For $E=\Pi k^2$, $\mathbb{S}_\lambda(\Pi k^2)=\mathbb{S}_{\lambda'}(k^2)$, and $\mathbb{S}_\mu(k^2)$ contains SL_2-invariants exactly for $\mu=(c,c)$. In degree $2d$ this selects $\lambda'=(d,d)$, i.e. $\lambda=(2^d)$, and $\mathbb{S}_{(d,d)}(k^2)=\det{}^d$ is one-dimensional. Hence

$$\big(A_{2d}^G\big)^{\mathrm{SL}_2}\cong L_d.$$

By External input 2.2(i), $A^G=A^{\mathfrak{g}}$ is generated by S for connected G, and S has degree 2. Under $(X,Y)\mapsto(g_{11}X+g_{12}Y,\ g_{21}X+g_{22}Y)$ with $g\in\mathrm{GL}_2$ one has $S\mapsto\det(g)\,S$, because $B(X,X)=B(Y,Y)=0$ and $B(Y,X)=-B(X,Y)$; so S is SL_2-invariant. Therefore $(A_{2d}^G)^{\mathrm{SL}_2}=k\,S^d$, and by (ii), (iii) its dimension is 1 for $d<h^\vee$ and 0 for $d\ge h^\vee$.

Even side. For $E=k^d$, $\mathbb{S}_\lambda(k^d)$ with $\lambda\vdash 2d$ contains SL_d-invariants exactly for $\lambda=(2^d)$, where $\mathbb{S}_{(2^d)}(k^d)=\det{}^2$. So the SL_d-invariant subspace of $k[\mathcal{C}_{\mathfrak{g}}^d]_{2d}^G$ is L_d. It contains the class of $\det T$, which is nonzero for $d<h^\vee$ by Corollary 3.3 and therefore spans it. □

5 The Cartan restriction and nilpotency

Lemma 5.1 (Semisimple parts; cf. [20], [16, §6], [8, §3])**.** *Let $\mathfrak{g}$ be semisimple and $(x_1,\dots,x_d)\in \mathcal{C}_{\mathfrak{g}}^d(k)$ with Jordan decompositions $x_i=s_i+n_i$. Then $s_1,\dots,s_d$ lie in one Cartan subalgebra and $f(x_1,\dots,x_d)=f(s_1,\dots,s_d)$ for every $f\in k[\mathfrak{g}^d]^G$.*

Proof. Each $\operatorname{ad} s_i$ and $\operatorname{ad} n_i$ is a polynomial in $\operatorname{ad} x_i$, so all s_i, n_j commute pairwise. The s_i span a toral subalgebra, contained in a Cartan subalgebra $\mathfrak{t}'$. Let $L=Z_G(s_1,\dots,s_d)^\circ$, a connected reductive subgroup that fixes each s_i, with Lie algebra $\mathfrak{l}=\mathfrak{z}_{\mathfrak{g}}(s_1,\dots,s_d)$. The commuting nilpotent elements $n_i\in\mathfrak{l}$ span an abelian subalgebra of nilpotent elements, which lies in the nilradical of a Borel subalgebra $\mathfrak{b}\subset\mathfrak{l}$. Let $B_L\supset T_L$ be a Borel subgroup and a maximal torus of L with $\operatorname{Lie} B_L=\mathfrak{b}$, and take a cocharacter μ of T_L with positive pairing with every root of the nilradical of $\mathfrak{b}$. Then $\mu(t)\cdot x_i=s_i+\mu(t)\cdot n_i\to s_i$ as $t\to 0$, and $f(x)=f(\mu(t)\cdot x)\to f(s)$. □

Proof of Corollary 2.4. By Corollary 3.3, the class of $\det T$ in $k[\mathcal{C}_{\mathfrak{g}}^d]^G$ is nonzero. Since $d>\dim\mathfrak{t}$, vectors $x_1,\dots,x_d\in\mathfrak{t}$ are linearly dependent, so $\Phi(\det T)=0$ (equivalently, Lemma 4.3 with $\lambda=(2^d)$). At every k-point of $\mathcal{C}_{\mathfrak{g}}^d$, Lemma 5.1 gives $\det T(x)=\det T(s)$ with $s_1,\dots,s_d$ in a Cartan subalgebra of dimension $\operatorname{rk}\mathfrak{g}<d$, so $\det T(x)=0$. By the Nullstellensatz, $\det T\in\sqrt{I_d}$, and its class is a nonzero nilpotent element of $k[\mathcal{C}_{\mathfrak{g}}^d]^G$. Conjecture 3.1 of [8] asserts that Φ is an isomorphism, and Conjecture 3.2 implies that Φ is injective [8, §3]. □

Remark 5.2. For $d\le\operatorname{rk}\mathfrak{g}$ (and then $d<h^\vee$), Φ maps the one-dimensional space L_d isomorphically onto $\mathbb{S}_{(2^d)}(\mathfrak{t}^*)^W$, which is one-dimensional. Indeed, the Gram determinant on $\mathfrak{t}^d$ is a nonzero element of $\mathbb{S}_{(2^d)}(\mathfrak{t}^*)^W$. Conversely, by Steinberg's theorem [2, Ch. V, §2, Exerc. 3] the W-module $\Lambda^d\mathfrak{t}^*$ is irreducible for $0\le d\le\operatorname{rk}\mathfrak{g}$; it is defined over $\mathbb{R}$, hence self-dual and orthogonal, so $\dim\operatorname{Sym}^2(\Lambda^d\mathfrak{t}^*)^W=1$. Since $\mathbb{S}_{(2^d)}(\mathfrak{t}^*)$ is a $\mathrm{GL}(\mathfrak{t}^*)$-summand of $\operatorname{Sym}^2(\Lambda^d\mathfrak{t}^*)$, $\dim\mathbb{S}_{(2^d)}(\mathfrak{t}^*)^W\le 1$.

6 Two examples

For a multilinear function m of six matrix arguments $(a_1,a_2,a_3,b_1,b_2,b_3)$ and $x_1,x_2,x_3\in\mathfrak{g}$ write

$$\langle m\rangle=\sum_{\sigma,\tau\in\mathfrak{S}_3}\operatorname{sgn}\sigma\,\operatorname{sgn}\tau\; m\big(x_{\sigma(1)},x_{\sigma(2)},x_{\sigma(3)},x_{\tau(1)},x_{\tau(2)},x_{\tau(3)}\big).$$

For a representation of $\mathfrak{g}$ with trace tr and B its trace form, $\langle\operatorname{tr}(a_1b_1)\operatorname{tr}(a_2b_2)\operatorname{tr}(a_3b_3)\rangle=6\det T$.

Example 6.1 ($\mathfrak{g}_2$, $d=3$)**.** Here $\operatorname{rk}\mathfrak{g}_2=2<3<4=h^\vee$. Let $Z_3=(\mathbb{S}_{(2,2,2)}\mathfrak{g}_2)^{G_2}$ be the SL_3-invariant part of $k[\mathfrak{g}_2^3]^{G_2}$ of degree 6, and $Y_3=(\mathfrak{g}_2\otimes\mathbb{S}_{(2,1,1)}\mathfrak{g}_2)^{G_2}$. Since $G_2\times\mathrm{SL}_3$ is reductive, $I_3\cap Z_3$ is the image of the $G_2\times\mathrm{SL}_3$-invariants of $\mathfrak{g}_2^*\otimes\Lambda^2k^3\otimes\operatorname{Sym}^4(k^3\otimes\mathfrak{g}_2^*)$ under multiplication by the commutator coordinates, and Pieri's rule ($\Lambda^2k^3\otimes\mathbb{S}_\mu k^3$ contains $\det^2$ iff $\mu=(2,1,1)$) identifies that space with Y_3. Exact character computations give

$$\dim Z_3=4,\qquad \dim Y_3=3,$$

so $Z_3/(I_3\cap Z_3)\neq 0$ by counting alone, Lemma 4.3 gives $\Phi(Z_3)=0$, and Theorem 2.3

identifies the quotient with $k \cdot \det T$. With traces in the 7-dimensional representation, let

$$S_{6a} = \langle \operatorname{tr}(a_1a_2a_3b_1b_2b_3)\rangle, \qquad S_{6b} = \langle \operatorname{tr}(a_1b_1a_2b_2a_3b_3)\rangle,$$
$$F_{42} = \langle \operatorname{tr}(a_1a_2b_1b_2)\operatorname{tr}(a_3b_3)\rangle, \qquad F_{33} = \langle \operatorname{tr}(a_1a_2a_3)\operatorname{tr}(b_1b_2b_3)\rangle.$$

Then Z_3 has basis $S_{6a}, S_{6b}, F_{42}, \det T$; $I_3 \cap Z_3$ has basis S_{6a}, S_{6b}, F_{42}; and $-18S_{6a} - 6S_{6b} + 3F_{42} + 4F_{33} = 0$ (Appendix A). For $\mathfrak{g}_2$ and $d = 3$, Φ is injective on every GL_3-isotypic component of degree at most 10 except $\mathbb{S}_{(2,2,2)}$, where its kernel is $k\cdot\det T$, and $(\det T)^2 \in I_3$ (Appendix A).
For comparison, $\mathfrak{sl}_3$, $\mathfrak{so}_5$ and $\mathfrak{sp}_4$ have $\mathrm{rk} = 2$ and $h^\vee = 3$, so Theorem 2.3 gives $\det T \in I_3$. Explicitly, with traces in the defining representation,

$$\begin{aligned}
\mathfrak{sl}_3:\ \det T = \Big\langle & -\tfrac{5}{6}\operatorname{tr}([a_1,a_2]a_3b_1b_2b_3) - \tfrac{2}{3}\operatorname{tr}([a_1,a_2]b_1a_3b_2b_3) \\
& + \tfrac{5}{18}\operatorname{tr}([a_1,a_2]a_3)\operatorname{tr}(b_1b_2b_3)\Big\rangle, \\
\mathfrak{sp}_4:\ \det T = \Big\langle & -\tfrac{1}{6}\operatorname{tr}([a_1,a_2]a_3b_1b_2b_3) - \tfrac{5}{6}\operatorname{tr}([a_1,a_2]b_1a_3b_2b_3) \\
& + \tfrac{1}{4}\operatorname{tr}([a_1,a_2]b_1b_3)\operatorname{tr}(a_3b_2)\Big\rangle, \\
\mathfrak{so}_5:\ \det T = \Big\langle & \tfrac{10}{3}\operatorname{tr}([a_1,a_2]a_3b_1b_2b_3) - \tfrac{4}{3}\operatorname{tr}([a_1,a_2]b_1a_3b_2b_3) \\
& - \tfrac{2}{3}\operatorname{tr}([a_1,a_2]a_3)\operatorname{tr}(b_1b_2b_3) + \operatorname{tr}([a_1,a_2]b_1b_3)\operatorname{tr}(a_3b_2)\Big\rangle.
\end{aligned}$$

Every term contains a commutator, so each right-hand side lies in I_3.

Example 6.2 ($\mathfrak{so}_7$, $d = 4$)**.** Here $\mathrm{rk} = 3 < 4 < 5 = h^\vee$, and Corollary 2.4 gives a nonzero nilpotent class $\det T \in k[\mathcal{C}^4_{\mathfrak{so}_7}]^{SO_7}$ with $\Phi(\det T) = 0$. The determinant $\det T$ is O_7-invariant, so the same element lies in $k[\mathcal{C}^4_{\mathfrak{so}_7}]^{O_7}$. In the SL_4-invariant part of degree 8,

$$\dim\big(\mathbb{S}_{(2,2,2,2)}\mathfrak{so}_7\big)^{SO_7} = 9, \qquad \dim\big(\mathfrak{so}_7 \otimes \mathbb{S}_{(2,2,1,1)}\mathfrak{so}_7\big)^{SO_7} = 12,$$

and the commutator substitution has rank at least 8 on the second space (rank 8 at random points modulo three primes; Appendix A). Theorem 2.3 gives rank exactly 8.

7 The orthogonal case of Song–Xia–Xu

For $G = O_n$ and for $G = SO_n$ with n odd, [23, Thm. 1.2] is [23, Thm. 3.4], deduced from [23, Prop. 3.1, Lemmas 3.2, 3.3]; for the orthogonal groups the degree bound [23, Lemma 3.3] is deduced in [23, §5.2] from [23, Lemma 5.3], whose proof invokes [23, Lemma 5.2] for the same d. For SO_{2m} the proof of [23, Thm. 4.3] uses [23, Thm. 3.4] for O_{2m}. The symplectic case rests on the Pfaffian of Chen–Ngô and is independent of [23, Lemma 5.2]. In this section (t_{ij}) is the generic skew-symmetric $(2d+3)\times(2d+3)$ matrix (called T in [23]), $h_{i_1\cdots i_k}$ is the Pfaffian of the matrix obtained by deleting the rows and columns $i_1, \ldots, i_k$ (a repeated index leaves a matrix of odd size, so $h_{i,i,\ldots} = 0$, consistent with $A^{\mathrm{SXX}}(1,1) = 2h^2_{1,3,4}$ in the case $d = 1$ of [23, §5.2]), and A^{SXX} is the $2d \times 2d$ matrix of [23, §5.2] (called A there),

$$A^{\mathrm{SXX}}(i,j) = h^2_{i,j,2d+2} + h^2_{i,2d+1,2d+2} + h^2_{j,2d+1,2d+2}, \qquad 1 \le i,j \le 2d.$$

[23, Lemma 5.2] asserts that $h_{2d+2} = 0$ and $h_{2d+3}\det A^{\mathrm{SXX}} \neq 0$ have a common solution.

Proposition 7.1. *On the hypersurface $h_{2d+2} = 0$, the matrix A^{SXX} has rank at most 3. Consequently $\det A^{\mathrm{SXX}} \in (h_{2d+2})$ for every $d \ge 2$, and [23, Lemma 5.2] fails for every $d \ge 2$.*

Proof. Let P be the skew-symmetric matrix of size $2d+2$ obtained from (t_{ij}) by deleting row and column $2d+2$, so $\operatorname{Pf} P = h_{2d+2}$; its rows and columns $1,\dots,2d+1$ are those of (t_{ij}), and its last one is the original $2d+3$. For $1 \le a < b \le 2d+2$ put $P^\sharp_{ab} = (-1)^{a+b} \operatorname{Pf} P(\hat{a},\hat{b})$ and $P^\sharp_{ba} = -P^\sharp_{ab}$. Expansion of the Pfaffian along a row gives $PP^\sharp = (\operatorname{Pf} P)\cdot \mathbf{1}$, and $h_{i,j,2d+2} = \pm P^\sharp_{ij}$, $h_{i,2d+1,2d+2} = \pm P^\sharp_{i,2d+1}$ for $i,j \le 2d$.
On $\operatorname{Pf} P = 0$ the columns of $P^\sharp$ lie in $\ker P$. If $\operatorname{rk} P = 2d$ this kernel has dimension 2; if $\operatorname{rk} P \le 2d-2$, every principal submatrix of size $2d$ is singular and $P^\sharp = 0$. In both cases the skew matrix $P^\sharp$ has rank at most 2, hence $P^\sharp = vw^{\mathsf{T}} - wv^{\mathsf{T}}$ for vectors v, w. Squares remove the signs:

$$h^2_{i,j,2d+2} = (v_i w_j - v_j w_i)^2, \qquad h^2_{i,2d+1,2d+2} = \ell_i^2, \quad \ell_i = w_{2d+1}v_i - v_{2d+1}w_i.$$

With $f_i = (v_i^2,\, w_i^2,\, v_i w_i,\, 1)$ and $a = w_{2d+1}^2$, $b = v_{2d+1}^2$, $c = -2w_{2d+1}v_{2d+1}$, expanding the squares gives $A^{\mathrm{SXX}}(i,j) = f_i M f_j^{\mathsf{T}}$ with

$$M = \begin{pmatrix} 0 & 1 & 0 & a \\ 1 & 0 & 0 & b \\ 0 & 0 & -2 & c \\ a & b & c & 0 \end{pmatrix}, \qquad \det M = -2\det \begin{pmatrix} 0 & 1 & a \\ 1 & 0 & b \\ a & b & 0 \end{pmatrix} + c^2 = -4ab + c^2 = 0.$$

Thus $\operatorname{rk} A^{\mathrm{SXX}} \le \operatorname{rk} M \le 3 < 2d$, and $\det A^{\mathrm{SXX}}$ vanishes at every k-point of $h_{2d+2} = 0$. The Pfaffian h_{2d+2} is irreducible [14, Lemma B.2.10], so (h_{2d+2}) is prime and the Nullstellensatz gives $\det A^{\mathrm{SXX}} \in (h_{2d+2})$. □

Remark 7.2. The proof uses polynomial identities over $\mathbb{Z}$ and linear algebra over a field of characteristic $\neq 2$. Hence [23, Lemma 5.2] also fails in characteristic $p > 2$ for $d \ge 2$, and the proofs of [23, Thms. 3.5, 4.4] (Theorem 1.2(2) there) for the orthogonal groups are incomplete for $d \ge 2$.

Remark 7.3 (The failing step)**.** The induction in the proof of [23, Lemma 5.2] (arXiv v3, p. 17) asserts that $\det A^{\mathrm{SXX}}$ has degree at most $4d-4$ in t_{12}, with coefficient of t_{12}^{4d-4} equal to

$$\det \begin{pmatrix} 2h^2_{1,2d+1,2d+2} & h^2_{1,2,2d+2} + h^2_{1,2d+1,2d+2} + h^2_{2,2d+1,2d+2} \\ h^2_{1,2,2d+2} + h^2_{1,2d+1,2d+2} + h^2_{2,2d+1,2d+2} & 2h^2_{2,2d+1,2d+2} \end{pmatrix} \cdot \det A',$$

with A' the matrix of the same form for the matrix obtained by deleting rows and columns $1,2$. For $j \ge 3$ the entries $A^{\mathrm{SXX}}(1,j)$ and $A^{\mathrm{SXX}}(2,j)$ contain $h^2_{j,2d+1,2d+2}$, which has degree 2 in t_{12}. The leading t_{12}^2-parts of rows 1 and 2 coincide, so the coefficient of t_{12}^{4d} vanishes; at $d=2$ and $d=3$ the exact degree is $4d-2$ (6 and 10), and at random integer values of the other t_{ij} the coefficient of t_{12}^{4d-4} differs from the displayed product (Appendix A). The base case $d=1$ is correct: there $\det A^{\mathrm{SXX}} = -(t_{35}^2 + (t_{25}+t_{15})^2)(t_{35}^2 + (t_{25}-t_{15})^2)$ (arXiv v3, p. 16), and Proposition 7.1 holds trivially because $2d = 2 \le 3$.

Corollary 7.4. *Let $n = 2m+1 \ge 7$ with $m+1 \le d \le 2m-2$, or $n = 2m \ge 8$ with $m+1 \le d \le 2m-3$. For $G = O_n$ and $G = SO_n$, with $W = N_G(\mathfrak{t})/Z_G(\mathfrak{t})$, the restriction $\Phi : k[\mathcal{C}^d_{\mathfrak{so}_n}]^G \to k[\mathfrak{t}^d]^W$ has a nonzero kernel and $k[\mathcal{C}^d_{\mathfrak{so}_n}]^G$ has a nonzero nilradical. In particular [23, Thm. 1.2(1), Cor. 1.3] fail for $n = 7$, $d = 4$. Since [23, Prop. 3.1, Lemma 3.2] are independent of [23, §5.2] and [23, Thm. 3.4] follows formally from them and [23, Lemma 3.3], the degree bound [23, Lemma 3.3] and [23, Lemma 5.3], which implies it, are false for $G = O_n$ and for $G = SO_n$ with n odd in the same range.*

Proof. These are the pairs (B_m, d) and (D_m, d) of Table 1. Corollary 2.4 gives the statement for SO_n; $\det T$ is O_n-invariant, so the same element works for O_n. □

8 Super-commuting schemes and $\mathcal{N} = 4$ Yang–Mills theory

Theorem 2.3 holds for every finite-dimensional super vector space E. For $E = k^{3|2}$ the scheme $\mathcal{C}_E = \mathcal{C}_{\mathfrak{g}}^{3|2}$ is the super-commuting scheme of [5, §2]. By Chang–Yin [7], the classical Q-cohomology of $\mathcal{N} = 4$ super-Yang–Mills theory with gauge algebra $\mathfrak{g}$ is the relative Lie algebra cohomology $H^\bullet(\mathfrak{g}[\mathsf{A}], \mathfrak{g}; \mathbb{C})$ with $\mathsf{A} = \mathbb{C}[z_+, z_-] \otimes \Lambda(\theta_1, \theta_2, \theta_3)$. By [5, Prop. 4], its part of top degree $p = |n|$ and multidegree $n = (n_{z_+}, n_{z_-}, n_{\theta_1}, n_{\theta_2}, n_{\theta_3})$ is $\mathcal{O}(\mathcal{C}_{\mathfrak{g}}^{3|2})_n^G$, the three even coordinates being $\partial_{\theta_i}\Psi|_{z=\theta=0}$ and the two odd ones $\partial_{z_\pm}\Psi|_{z=\theta=0}$ for the BPS superfield Ψ of [5, §3]. Following [5, Def. 1], a non-Cartan class is a nonzero element of $\ker \Phi_{k^{3|2}}$; the level of n is $3n_{z_+} + 3n_{z_-} + 2(n_{\theta_1} + n_{\theta_2} + n_{\theta_3})$.

Corollary 8.1. *Let $\mathfrak{g}$ be simple and $1 \le d < h^\vee$. With respect to the Borel subalgebra of $\mathfrak{gl}(3|2)$ with the even basis vectors first, $\mathbb{S}_{(2^d)}(k^{3|2})$ is nonzero with highest weight*

$$\big((2^{\min(d,3)}, 0^{3-\min(d,3)}) \,\big|\, (c, c)\big), \qquad c = \max(d-3, 0).$$

Hence $\mathcal{O}(\mathcal{C}_{\mathfrak{g}}^{3|2})^G$ contains $\mathbb{S}_{(2^d)}(k^{3|2}) \otimes L_d \neq 0$, of degree $2d$; its highest-weight vector has multidegree

$$n = (0,0,2,0,0),\ (0,0,2,2,0),\ (d-3,d-3,2,2,2) \qquad (d=1,\ d=2,\ d\ge 3), \qquad p = |n| = 2d,$$

of level $4d + 2c$. Its part of multidegree $n = (d, d, 0, 0, 0)$ is $k\,S^d$, with $S = B(X, Y)$ for the two odd coordinates $X = \partial_{z_+}\Psi|_{z=\theta=0}$, $Y = \partial_{z_-}\Psi|_{z=\theta=0}$. For $d > \operatorname{rk}\mathfrak{g}$ this component lies in $\ker \Phi_{k^{3|2}}$.

Proof. The hook condition $\lambda_4 \le 2$ of External input 4.2 holds for $\lambda = (2^d)$. For the distinguished Borel subalgebra the highest weight of $\mathbb{S}_\lambda(k^{3|2})$ is $\big(\lambda_1, \lambda_2, \lambda_3 \,\big|\, \max(\lambda_1' - 3, 0), \max(\lambda_2' - 3, 0)\big)$ [1], which for $\lambda = (2^d)$ is the stated weight. Write $k^{3|2} = k^3 \oplus \Pi k^2$. In Lemma 4.1 the part of $E^{\otimes n}$ of degree zero in k^3 is $(\Pi k^2)^{\otimes n}$, so the part of $\mathcal{O}(\mathcal{C}_{k^{3|2}})$ of degree zero in the even coordinates is $\mathcal{O}(\mathcal{C}_{\Pi k^2}) = A$, and the corresponding part of $\mathbb{S}_{(2^d)}(k^{3|2})$ is $\mathbb{S}_{(2^d)}(\Pi k^2) = \det{}^d$, of multidegree $(d, d, 0, 0, 0)$. By the odd side of the proof of Theorem 2.3, the part of the component of multidegree $(d, d, 0, 0, 0)$ is $(A_{2d}^G)^{\mathrm{SL}_2} = k\,S^d$. The rest follows from Theorem 2.3 and Lemma 4.3. □

Corollary 8.2. *If Φ_{k^d} is an isomorphism for every $d \ge 1$, then Φ_E is an isomorphism for every finite-dimensional super vector space E. In particular, for $\mathfrak{g} = \mathfrak{sp}_{2N}$ the map $\Phi_{k^{3|2}}$ is an isomorphism, and every nonzero element of $\mathcal{O}(\mathcal{C}_{\mathfrak{sp}_{2N}}^{3|2})^G$ has nonzero Cartan restriction.*

Proof. By Lemma 4.3, $\Phi_{k^d} = \bigoplus_\lambda \mathrm{id} \otimes \phi_\lambda$ with $\mathbb{S}_\lambda(k^d) \neq 0$ for every λ with at most d parts; so all ϕ_λ are isomorphisms, and $\Phi_E = \bigoplus_\lambda \mathrm{id} \otimes \phi_\lambda$ is an isomorphism. For $\mathfrak{sp}_{2N}$ the hypothesis is [9, Thm. 4.1]. □

Remark 8.3. (a) For $\mathfrak{g}_2$ ($\mathrm{rk} = 2$, $h^\vee = 4$), $d = 3$ gives the non-Cartan class of $\det T$ at $n = (0, 0, 2, 2, 2)$, of level 12; in the conventions of [5, §5.2] its charges are $(J_1, J_2, q_1, q_2, q_3) = (0, 0, 2, 2, 2)$. It is a purely scalar element of the classical Q-cohomology.

(b) For $\mathfrak{so}_7$ (rk $= 3$, $h^\vee = 5$), $d = 4$ gives a non-Cartan class at $n = (1,1,2,2,2)$, $p = 8$, of level 18. By [6, §3, Table 1 and Discussions], $\mathcal{O}(\mathcal{C}^{3|2}_{\mathfrak{so}_7})^{SO_7}_n$ is 85-dimensional there (84 for $\mathfrak{sp}_6$), and super-abelianization, which is $\Phi_{k^{3|2}}$, has a one-dimensional kernel on it. The class of Corollary 8.1 therefore spans this kernel, and the non-Cartan class of [5, Thm. 1] is a nonzero multiple of it. A representative of that class, due to Choi and Lee [10] and written in superfield form in [5, eq. (16)], is $\varepsilon_{ijk}\varepsilon_{lmn}Z_{il}\operatorname{Tr}(\Psi_j\Psi_m)\operatorname{Tr}(\Psi_k\Psi_n)$ with $Z_{il} = \operatorname{Tr}(\Psi_i\Psi_l)\operatorname{Tr}(\Psi_{\dot\alpha}\Psi^{\dot\alpha}) - \operatorname{Tr}(\Psi_i\Psi_{\dot\alpha})\operatorname{Tr}(\Psi_l\Psi^{\dot\alpha}) - 4\operatorname{Tr}(\Psi_{\dot\alpha}\Psi^{\dot\alpha}\Psi_{(i}\Psi_{l)})$; in the notation of [5], Ψ_i and $\Psi_{\dot\alpha}$ are the even and odd coordinates.

(c) The Langlands-dual pair $(\mathfrak{so}_{2N+1}, \mathfrak{sp}_{2N})$ has rank N and dual Coxeter numbers $2N-1$ and $N+1$. Corollary 8.1 gives non-Cartan classes for $\mathfrak{so}_{2N+1}$ at $n = (d-3, d-3, 2, 2, 2)$ for $N+1 \le d \le 2N-2$, and by Corollary 8.2 the Cartan restriction is injective for $\mathfrak{sp}_{2N}$. For $N = 3$ this is the non-Cartan part of the classical mismatch at level 18 in [6]; for $N = 4$ it gives non-Cartan $\mathfrak{so}_9$ classes at $n = (2,2,2,2,2)$ and $(3,3,2,2,2)$.

(d) In the notation of [17, §2] the odd coordinates X, Y are the gaugino components $\lambda_\pm$, and S is a nonzero multiple of $\operatorname{Tr}(\lambda_+\lambda_-)$. For $\operatorname{rk}\mathfrak{g} < d < h^\vee$ the vector S^d of Corollary 8.1 is the non-Coulomb monotone class $[S^d]$ of [17, eq. (1)]. Corollary 8.1 places S^d and the class at $n = (d-3, d-3, 2, 2, 2)$ in one irreducible $\mathfrak{gl}(3|2)$-module. For $\mathfrak{so}_7$, $d = 4$, this agrees with [17, §5, App. D], where S^4 is shown to be a superconformal descendant of the class in (b).

A Computations

Exact counts come from Weyl characters (two independent character routes for the $d = 3$ rows). Ranks are ranks of matrices of function values at random points: such a rank is a lower bound for the rank over $\mathbb{Q}$, and it is exact when it equals the dimension of the source or the target. The $d = 3$ ranks were also computed exactly over $\mathbb{Z}$, with Example 6.1 recomputed in a second realization of $\mathfrak{g}_2$ (the stabilizer in $\mathfrak{so}_7$ of the octonionic 3-form).

$\mathfrak{g}$, d	$(\dim Y_d, \dim Z_d)$	rank	$\det T \in I_d$	$\dim Z_d - \text{rank}$	$\dim L_d$	$\Phi(\det T)$
$\mathfrak{sl}_3$, 3	$(4,3)$	3	yes	0	0	–
$\mathfrak{so}_5, \mathfrak{sp}_4$, 3	$(4,4)$	4	yes	0	0	–
$\mathfrak{g}_2$, 3	$(3,4)$	3	no	1	1	0
$\mathfrak{so}_7, \mathfrak{sp}_6$, 3	$(4,5)$	4	no	1	1	$\neq 0$
$\mathfrak{sl}_4$, 3	$(5,5)$	≥ 4	no	≤ 1	1	$\neq 0$
$\mathfrak{so}_5$, 4	$(9,5)$	5	yes	0	0	–
$\mathfrak{g}_2$, 4	$(7,5)$	5	yes	0	0	–
$\mathfrak{sl}_4$, 4	$(20,8)$	8	yes	0	0	–
$\mathfrak{sp}_6$, 4	$(12,8)$	8	yes	0	0	–
$\mathfrak{so}_7$, 4	$(12,9)$	≥ 8	no	≤ 1	1	0

Table 2: $Z_d = (\mathbb{S}_{(2^d)}\mathfrak{g})^G$ and $Y_d = (\mathfrak{g}\otimes\mathbb{S}_{(2^{d-2},1,1)}\mathfrak{g})^G$, the source of the commutator substitution; "rank" is the rank of $Y_d \to Z_d$ and $\dim L_d$ is given by Theorem 2.3. Entries "no" in the column $\det T \in I_d$ follow from Corollary 3.3 for the rows where the computed rank is a lower bound ($\mathfrak{sl}_4$, $d = 3$ and $\mathfrak{so}_7$, $d = 4$) and from the exact computation in the other rows.

- $\mathfrak{g}_2$, $d = 3$: the 265 contractions $\langle m\rangle$ of fixed-point-free multilinear trace monomials (in the 7-dimensional representation) have rank $4 = \dim Z_3$ over $\mathbb{Z}$ at 16 random integer points; adding 150 elements of Z_3 built from the 3-form φ and the 4-form $*\varphi$ leaves the rank at 4. The contractions $\langle m\rangle$ of five-slot monomials with a commutator in one slot span a space of rank $3 = \dim Y_3$, containing $S_{6a}, S_{6b}, F_{42}, F_{33}$; adding $\det T$ raises the rank to 4.

- $\mathfrak{g}_2$, degree 12: the SL_3-invariant part of $k[\mathfrak{g}_2^3]^{G_2}$ of degree 12 is $(\mathbb{S}_{(4,4,4)}\mathfrak{g}_2)^{G_2}$, of dimension 30; the commutator image has rank 30 modulo 10^9+7, so $(\det T)^2 \in I_3$.
- $\mathfrak{g}_2$, $d=3$, degree at most 10: for every partition λ with $|\lambda| \le 10$ and at most 3 parts, Φ is injective on the $\mathbb{S}_\lambda$-isotypic component of $k[\mathcal{C}^3_{\mathfrak{g}_2}]^{G_2}$, except for $\lambda=(2,2,2)$.
- Lemma 3.1: checked symbolically for $d \le 5$.
- Proposition 7.1: at random rational points of $h_{2d+2}=0$, $\operatorname{rk} A^{\mathrm{SXX}}=2$ and $\det A^{\mathrm{SXX}} \neq 0$ for $d=1$; $\operatorname{rk} A^{\mathrm{SXX}}=3$ and $\det A^{\mathrm{SXX}}=0$ for $d=2,3,4$. Remark 7.3: at $d=2$ and $d=3$, $\deg_{t_{12}} \det A^{\mathrm{SXX}} = 4d-2$, and the coefficient of t_{12}^{4d-4} differs from the product in Remark 7.3 at three random integer points.

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